

Ali Alemami

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Systems and backend developer with hands-on experience building Unix shells, HTTP servers, and concurrent simulations in C/C++ at 42 Amman, alongside enterprise desktop applications in C#.NET. CS student at Hashemite University, 22/24 projects into the 42 Common Core.

EDUCATION

42 Amman Common Core, 22/24 Projects Complete	2025 - Present
- Peer-to-peer software engineering curriculum: C/C++, POSIX APIs, memory management, concurrency. - Conducted 50+ peer code reviews and defense audits, evaluating memory leaks (Valgrind), race conditions, and architecture.	
The Hashemite University BSc Computer Science	2022 - 2026 (Expected)
- Coursework: Algorithms, Data Structures, Computer Networks, Operating Systems.	

TECHNICAL SKILLS

Languages: C (POSIX/C99), C++ (98/11/17), C# (.NET), SQL (T-SQL), Bash

Systems

42 CORE PROJECTS

Philosophers (github.com/ali-alemami/philos)
Race-free multithreaded Dining Philosophers simulation using pthreads, mutex locks, async monitors, and microsecond delta timers.
Webserv (github.com/ali-alemami/webserv)
Non-blocking HTTP/1.1 server using epoll I/O multiplexing, stateful request parsing (GET/POST/DELETE), NGINX-style configs, and CGI execution.
Inception (github.com/ali-alemami/inception)
Multi-service Docker Compose infrastructure: NGINX (TLS v1.2/v1.3), WordPress (PHP-FPM), MariaDB on Alpine Linux.
Minishell (github.com/ali-alemami/minishell)
POSIX-compliant Unix shell with AST parser, subshell forks, pipelines, heredocs, variable expansion, and signal handling; zero leaks.
mini_rt (github.com/ali-alemami/mini_rt)
3D raytracing engine from scratch: vector math, Phong illumination (diffuse/ambient/specular/shadows), geometric intersections.
C++ Modules 00-09 (github.com/ali-alemami/cpps)
OOP, Orthodox Canonical Form, polymorphism, templates, STL; Ford-Johnson Merge-Insert Sort with Jacobsthal pairing heuristics.
push_swap (github.com/ali-alemami/push_swap)
Two-stack sorting via bitwise radix operations optimized for minimal instruction counts.

SOFTWARE DEVELOPMENT & DATABASE PROJECTS

Bank & ATM System (github.com/ali-alemami/Bank-and-ATM-System-CPP)
Dual-engine C++ banking app with bitmask access controls, custom file serialization, transaction audit logs, and file persistence.
Contacts Management App (github.com/ali-alemami/Contacts-App-3Tier)
3-tier C#.NET desktop app (Presentation / Business / Data Access layers), parameterized T-SQL via ADO.NET, SQL Server backend.
Algorithms & Problem Solving (github.com/ali-alemami/Programming-Advices-Core)
225+ solutions across 5 progressive levels: data structures, matrix ops, string parsing, combinatorics, date/time arithmetic.
Database Schemas & SQL (github.com/ali-alemami/Programming-Advices-Core)
5 relational schemas (Medical Clinic, E-Commerce, HR/Payroll, Vehicle Registry, Employee Analytics) with analytical JOINS and window functions.
WinForms Collection (github.com/ali-alemami/CSharp-WinForms-Collection)
10+ desktop utility and game applications in C# focusing on UI event handling, input validation, and OOP principles.